SoK: When Do LLM Side-Channel Defenses Protect Agent Workflows?

***Abstract*—When does a defense for an LLM service protect the agent workflow built on it? We answer this question through result-level synthesis: each measurement is linked to its receiver, execution boundary and evaluated condition. Three checks turn heterogeneous findings into deployment decisions. A content-aware frontend can regenerate a signal after upstream protection; cover that enters action execution requires task-state validation; and adaptation effects and overheads apply to the mechanisms actually evaluated. Worked cases contrast downstream re-emission with preserved features, privacy-oriented decoy execution with state-changing humanization, and component prices with service costs. The contribution is a decision procedure supported by source-linked cases and a reusable map of eight defense types, grounded in a 60-family collection and 33 scoped result records. Its positive output is explicit reuse conditions: retain matched-view evidence, verify goal-neutral cover, and bind protection and cost to the same mechanism. This makes existing measurements actionable without forcing them into a misleading defense ranking.**

1. Introduction

Encryption conceals content but need not conceal execution. LLM services produce token streams, cache hit/miss timing and shared-resource activity; agents add destinations, tool exchanges and website actions. A receiving website, a passive network observer and an active API client see different projections [1], [2], [3], [4], [5], [6]. The practical question is which deployment decision a favorable defense result actually supports.

A designer choosing padding, cover traffic or action humanization faces a concrete difficulty: similar defense labels refer to different protected objects. NetEcho studies downstream re-emission, the Web and Research Agents (WRA) study uses concurrent decoy requests for privacy, Dangol measures transport cover, and the Agent Humanization Benchmark (AHB) measures executed mobile humanization [7], [6], [8], [9]. A lower inference score, preserved task success and a small component overhead are useful findings, but they answer different deployment questions. Reading them as one defense ranking loses the conditions that make each result useful.

Central insight: follow both the trace and the action. Protecting a communication trace and preserving an agent's task are separate obligations. A downstream frontend can emit a new target-bearing trace after upstream protection; a dummy that enters tool or GUI execution can alter the task itself. The distinction changes the appropriate evaluation. A new emitter calls for final-hop capture; surviving features call for an adapted predictor; executed cover calls for objective task and policy checks. These tests address different failure mechanisms and cannot substitute for one another.

We turn these source findings into three worked deployment decisions (Section 5): choose the final interface to protect, decide whether cover may execute, and select an applicable adaptation result and cost. Each case states what can be adopted from existing evidence and what condition would permit reuse elsewhere. For example, verified discard before semantic dispatch excludes direct fake-action state changes while leaving resource and delay effects to be checked. The same tested receiving view supports its local adaptation result; a different view requires a capture bridge. This cross-paper mapping is the contribution: the original papers establish the component phenomena, and the synthesis makes their consequences usable for system design.

Contributions and use. The paper contributes (1) three source-backed deployment decisions with explicit conditions for reusing a result; (2) an eight-type defense map and a quantitative audit of result comparability and reporting; and (3) source-linked records that make these judgments reconstructible. The October 8, 2026 collection contains 60 families: 43 core (15 agent-native and 28 runtime/application-inherited) plus 17 adjacent. This collection supplies the context and counterconditions; the individual measurements supply each case's evidence. Privileged diagnostics, attribution and human–agent distinction are distinguished from confidentiality leakage.

Figure 1 separates the trace and action paths; Figure 2 gives the three-check procedure. Section 5 works through its cases, and Section 6 audits the evidence. The checklist in Section 8 translates the cases into a reporting practice. Public-trace reanalysis, failed runs and the negative checklist comparison are retained in the Supplementary Evidence, where they illustrate the scope of particular judgments rather than carry the deployment claims.

2. Scope and Review Method

2.1. Questions and eligibility

We ask which hidden targets and observables appear (RQ1), how signal origin relates to receiver permission (RQ2), what establishes feasibility and transfer (RQ3), and which intervention addresses which observable at what cost

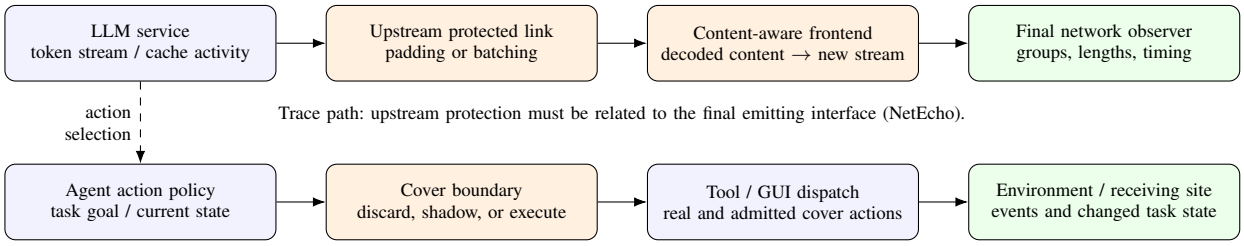


Figure 1. Two paths that a deployment judgment must follow. The top path distinguishes an upstream protected link from a new downstream emitter; the bottom distinguishes communication cover from actions admitted to execution. The diagram organizes source mechanisms, not one jointly evaluated system.

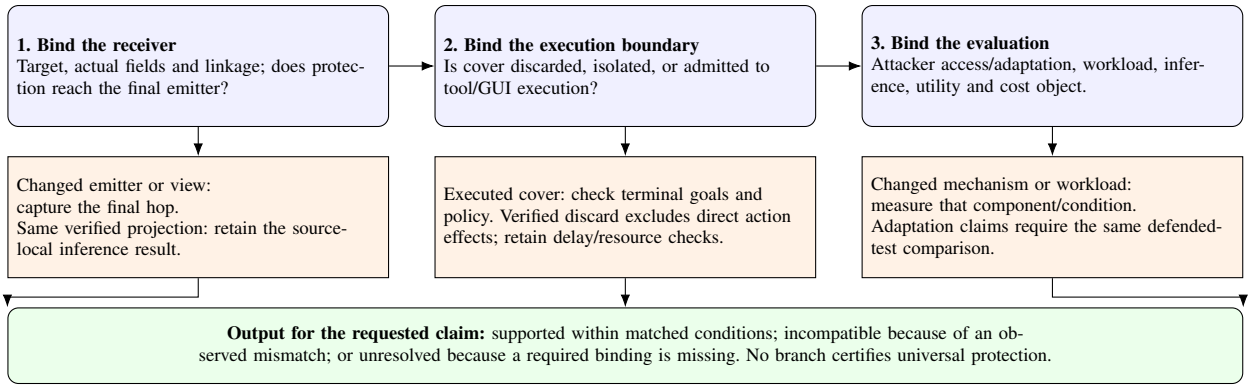


Figure 2. Three checks for applying a reported defense result. Each identifies a positive reuse condition and the evidence needed when the deployment changes.

(RQ4). An agent selects and executes actions using an LLM, tools, memory or orchestration; inherited serving and hardware channels need not be agent-specific.

Core eligibility requires (C1) LLM/agent execution, (C2) a hidden target, (C3) an execution-dependent auxiliary observable rather than ordinary output semantics alone, (C4) specified observation permissions, and (C5) an empirical inference evaluation. A controlled synthetic-state inference can satisfy C5; a proposed mechanism or timing gap alone cannot. Mixed papers qualify through an identifiable component. Table 1 supplies the decision rules.

The final coded collection contains 60 families available by October 8, 2026: 43 core (15 workflow-origin, 28 inherited-runtime/application) and 17 adjacent. Four core components concern authorized diagnostics. Other targets include content/property inference, provenance attribution and human/agent distinction; hiddenness alone establishes no confidentiality violation. Families are deduplicated studies; result records distinguish conditions within a study.

The auxiliary-only rule is applied consistently. Coding fingerprints and WildAuth remain adjacent because their evaluated views mix semantic and interaction features; feature importance is not a separately evaluated auxiliary-only attack [10], [11]. Metric-only fault localization qualifies as a diagnostic component despite operator authorization [12]. ZeroPrint's response differences, endpoint-content fingerprints and StepFinder's mixed view remain adjacent [13], [14], [15].

TABLE 1. OBSERVABLE-BASED ELIGIBILITY. A HIDDEN TARGET ALONE DOES NOT MAKE A SEMANTIC ATTACK A SIDE CHANNEL.

Observed input	Decision and required distinction
Length, timing, cache response	Core when measured execution dependence supports hidden-target inference.
Own-site actions	Core attribution with site permission; no automatic access to other sites or private tool logs.
Reply states the secret	Direct disclosure; adjacent, not auxiliary-channel evidence.
Reply style/content identifies a user	Adjacent unless a separately evaluated auxiliary mechanism is established. WildAuth stays a mixed-view comparator.
Output-filter membership oracle	Can be core: the filter's execution, not ordinary semantic output inference, mediates the signal.
Privileged loop/fault logs	Separate diagnostic component when C1–C5 hold; does not increase confidentiality evidence.

2.2. Purposive sampling and citation tracing

This SoK uses purposive, mechanism-oriented sampling supplemented by fixed one-hop backward and forward

citation tracing. It synthesizes decision-relevant evidence rather than estimating literature-wide prevalence. Acquisition combines native IEEE Xplore/ACM Digital Library exports, targeted discovery of preprints/proceedings with exact-title/DOI primary-source lookup, and tracing from 40 fixed seeds. Targeted searches covered arXiv and primary USENIX, NDSS and PMLR records. Appendix A gives executable native queries; Supplement S1 retains their exports and targeted query logs. All 2,237 returned backward reference occurrences and 24 distinct forward works were screened; forward lookup resolved 38 of 40 seeds. Versions are merged only with identity evidence.

We distinguish the acquired candidate frames from the coded collection. The prepared metadata frame contains 4,753 unique IDs with a recorded exclusion, referral or unresolved disposition. A bounded source reconciliation accounts for 109 overlapping decision rows in four nominated frames (57 follow-ups, six proposals, 22 backward candidates and 24 forward records). The map retains source-unavailable outcomes; it is not 109 independent full readings. At the source-search endpoint, unavailable full texts and unestablished empirical components are not admitted to the present core analysis. Unbound defense quantities are excluded from matched comparisons without removing their qualifying attack families.

The review endpoint is the named, version-bound 60-family collection. The saved native exports contain only 1/24 (IEEE) and 0/24 (ACM) fixed known papers. They are therefore inadequate as the sole acquisition route; targeted primary-source discovery and citation tracing are essential parts of this purposive design. Because database indexing and field eligibility are not established for all 24, these figures measure export presence rather than a calibrated recall rate. The unacquired broad-query frame and unresolved routes are retained in Supplement S1. Countercondition searches and source-deletion analysis test the specific synthesis, without converting it into an exhaustive systematic review.

2.3. Coding, adjudication and reproducibility

The extraction dictionary specifies receiver permissions, auxiliary versus semantic observations, result splitting, execution mode, utility/cost object and source/run binding. Source-scoped NR-R, unavailable NR-A and unverified NR-V remain distinct from replication. Code inspection can establish implementation behavior without authenticating the historical run.

One coding workflow used AI assistance in screening, extraction, source rechecks and writing. Disagreements were resolved against source text and recorded with locators. Supplement S1 retains the 23-case blinded AI exercise (17/21 determinate agreement), 332-dispute readbacks and exclusion audits. They are AI-assisted checks, not independent human coding or a human agreement estimate. The result records and conclusion dependencies expose the interpretations for direct inspection.

TABLE 2. A SINGLE RESULT RECORD. ONLY FIELDS RELEVANT TO THE REQUESTED COMPARISON MUST ALIGN; UNAVAILABLE REQUIRED FIELDS REMAIN UNRESOLVED.

Record field	Required meaning
Source	Family/result ID, version, source locator, reading scope and hash.
Target	Hidden property, inference goal and protection relation: confidentiality, attribution, human/agent distinction or authorized diagnosis.
Receiver	Actual permission and released representation, including session/document linkage and any auxiliary inputs.
Intervention	Where protection ends; whether edits affect stored traces, transmission or executed actions.
Attacker	Fit/calibration condition, test population, training access and adaptation budget.
Workload	Task, system, topology/hardware and grouping/holdout.
Outcomes	Inference metric/denominator; objective or proxy utility; applicable cost object; uncertainty and field states.

All 43 core families have a scoped intervention disposition. The Supplementary Evidence supplies the 60-family index, 33 result records, 66 utility/cost field positions, source versions/locators, claim links and a fit-free verifier. Section 3 gives a complete record-to-judgment example. Supplement S6 tests source deletion and unresolved bindings; S8 supplies the pairwise and checklist audit with its coding rules and row-level reasons.

3. Result Records and Comparison Rules

The unit of comparison is one evaluated condition, rather than a paper or a headline score. A result record binds a source version and locator to its target, receiver, intervention, attacker procedure, workload and outcomes (Table 2). A family can contribute several records when it reports different execution modes, protected views or cost components. This is why a valid result can answer one deployment question while leaving another unresolved.

Fields, not a competing framework. The mnemonic SCOPE expands to Secret/target, Channel, Observer, Pipeline and Exploitation goal. It is a record-field schema. Source/evaluation columns, historically called the Evidence Plane, supply provenance and experimental conditions for the same record. Generating layers span hardware, serving, transport, application/API, orchestration, tools/MCP, environment and multi-agent communication; they can differ from the receiving layer. Permission, activity, calibration access and protection relation remain separate fields. Appendix A provides the compact key used by the detailed tables.

Three comparison outcomes. A *compatible* comparison has evidence for the fields needed by the requested inference. An *incompatible* comparison has a positively established mismatch, such as token payload versus wire bytes or stored timestamps versus executed actions. An *unresolved* comparison lacks a necessary binding or transfer

TABLE 3. COMPARISON RULES. A KNOWN MISMATCH AND A MISSING BINDING HAVE DIFFERENT CONSEQUENCES.

Requirement	Condition and resulting license
Descriptive comparison	Align target, receiver permission, metric meaning and cost object. Different systems may be described together; token payload and wire bytes cannot be ranked as the same cost.
Paired effect	Compare the same defended test units and outcome. A known test-unit mismatch makes that subtraction incompatible; an unverified fit binding leaves the effect unresolved.
Deployment transfer	Establish a bridge for the changed receiver, intervention, workload, topology or hardware. A demonstrated violation blocks the proposed transfer; an untested bridge leaves it unresolved. Permit only the range and endpoints actually covered.

test. Only the affected inference is withheld: an unreported task endpoint does not invalidate a measured bandwidth cost. Table 3 applies these outcomes to descriptive comparison, paired effects and deployment transfer.

Missingness and adaptation. NR-R records a quantity unreported within the inspected source scope; NR-A records unavailable material; NR-V records an unverified field, version or run binding; NA records inapplicability. A missing field gives an unresolved judgment, not an observed mismatch. Fitting on protected data (D1) and comparing clean- versus protected-trained procedures on the same defended test population (D2) are separate entries in the attacker column. Neither alone establishes that the observer can obtain the labeled data or afford the fitting budget.

A complete traceable example. Table 4 connects a family, a result, the inspected source and the deployment judgment. The same identifiers join the supplied family index, result ledger and claim links; full source hashes and reading scopes remain in the companion evidence package.

Positioning the schema. The contribution is the source-linked decisions produced from these fields, not an advantage of the mnemonic over a checklist. A retained 512-root constructed-task comparison gives 32.16% joint correctness for receiver traversal versus 40.63% for a flat checklist and 45.12–45.31% for a structured checklist. We therefore use the field schema for indexing only. Supplement S7 preserves the original endpoint, negative comparison and missing-score treatment.

4. Corpus Map and Nearest Prior Work

4.1. Origins and receiving projections

Runtime/application-inherited signals arise from inference, serving and application mechanisms; agent-native signals arise from orchestration, tools, browser actions and multi-agent execution. Origin is not receiver location. Supplement S2 maps all source-coded origins and the load-bearing source–receiver–intervention paths, distinguishing

TABLE 4. FILLED RECORD AND ITS LICENSED JUDGMENT. VALUES ARE SOURCE-REPORTED; NO CROSS-MODE EFFECT IS COMPUTED.

Join key / field	Entry
Family / result	`dangol2026workflow` / `DANGOL-livebinding`.
Version / locator	arXiv:2606.07150v3, §9.7, live-protocol paragraph after Table 5 [8].
Target / view	Workflow classification from live Oblivious HTTP (OHTTP)/Agent-to-Agent (A2A) traffic; not GUI-state observation.
Intervention	Live fixed-rate cover with fresh per-session identifiers, distinct from transformed stored captures.
Workload / testbed	Workflow classes with client, relay, gateway and agent co-located on one loopback host.
Outcome / cost	Accuracy 0.30; about 13× bytes and 11× latency in that condition.
Unresolved fields	Objective terminal task utility NR-R; precise paired live-refit comparison NR-V.
Licensed judgment	These costs describe this live protocol condition. They neither price dummy GUI actions nor prove task preservation. A matched objective endpoint is needed for the latter.

diagnostic components from confidentiality claims. These maps locate relevant evidence; the three cases make the deployment judgments.

Inherited channels. Weiss reconstructs candidate text from encrypted length sequences; NetEcho's semantic information recovery is not exact-text recovery [1], [7]. Whisper Leak classifies topics from traffic, while Time Will Tell uses aggregate completion time [16], [17]. Different targets and metrics prevent treating their percentages as attack strength. Chosen-request cache attacks use shared-state timing oracles; local CPU-cache and GPU interference attacks require different permissions [2], [3], [18], [19], [20], [21]. LeakyLMs' architecture branch requires same-GPU-class calibration; KVGov separates hardware timing, simulated inference/defense and analytic controls [22], [23].

Workflow channels and target relations. AgentPrint, the Web and Research Agents (WRA) study, and the traffic benchmark infer identities, activities or traits through flow/destination structure. UI Traces and FP-Agent use own-site behavior or browser attributes [4], [6], [24], [5], [25]. URL and domain views are distinct; passive encrypted observation does not reveal endpoint method semantics. AgentTell's probe site, PlanTwin's planner and SICC's authorized recipient occupy still different projections [26], [27], [28].

The 43 core families include 39 non-diagnostic components and four privileged diagnostics (ClawGuard, cycle detection, LumiMAS and LLM-serving fault localization). The latter uses 120 controlled fault injections and supplied onset times, not a low-permission privacy observer [12]. Model/human attribution is not by itself a confidentiality violation. Coding fingerprints, WildAuth, SymDiag, ZeroPrint, endpoint fingerprints and StepFinder are adjacent semantic-boundary comparisons. AgentDAM is a background data-minimization benchmark, not auxiliary inference [10], [11], [29], [30]. The target-relation ledger records these differences without converting every hidden property

into a protected secret.

4.2. What earlier systematizations already establish

Prior systematizations establish cross-layer composition and the scope of inference-to-agent transfer. Banerjee et al. organize compound-system attacks by cross-layer mechanisms and MITRE threat models; Forough et al. v2 distinguish transferable inference protection from memory/action/coordination requirements under confidential computing [31], [32]. We build on these perspectives by making the unit of comparison a measured result and the output a deployment decision. These prior-systematization sources supply background rather than C5 attack families.

Debenedetti's auxiliary-component privacy channel, also cited by Banerjee, supplies a shared composition precursor. Our increment is the result-level decision: which existing measurement licenses a proposed deployment, and which bridge would make another result applicable? NetEcho motivates final-hop measurement after upstream protection; WRA/AHB distinguish a privacy-oriented reason to execute decoys from their possible task-state effects. Likewise, Forough distinguishes a trusted execution environment (TEE) boundary from side-channel resistance; the Early Bird/KVGov comparison makes the consequence concrete through matching primitives, calibrated oracles and measured cost components. Section 5 turns these distinctions into permitted uses of the existing evidence.

Classical website-fingerprinting defenses establish traffic–cost trade-offs: Tamaraw studies shaping and bandwidth bounds, WTF-PAD uses adaptive padding, FRONT randomizes padding near the trace front, and Walkie-Talkie combines half-duplex communication with burst molding [33], [34], [35], [36]. The agent-specific comparison follows what happens after protection terminates: a content-aware frontend can emit another signal, and a cover action can enter task execution. Lahjouji–Colaco address agent data surfaces and governance [37]; Kim and Horváth et al. supply complementary agent-security and physical-channel maps [38], [39]. Supplement S2 gives the full prior-work comparison. Our added artifact is a result-to-decision map, not another claim to have discovered cross-layer composition.

5. Three Deployment Decisions

The three checks in Figure 2 answer concrete design decisions. Table 5 collects the resulting permissions and boundaries so that each case can focus on what changes in the deployment judgment.

5.1. Case 1: choose the final interface to protect

Decision and relevant evidence. An application designer must decide whether upstream API padding or batching also protects the user-facing connection after a content-aware frontend. NetEcho identifies the relevant receiving path: Trace A is token counts per group; Trace B estimates token/group character lengths. A downstream application can consume decoded upstream content or stream state and emit finer-grained network observations [7]. Captured deployment configurations and locally implemented active-defense settings are distinct source results. Neither exact text nor original generation timestamps are automatically recovered.

What changes after binding. An upstream padded link answers a different question from protection of the final emitter. NetEcho's authors already establish re-emission in their systems. The joint synthesis specifies the next measurement: capture the last target-bearing interface rather than merely refit on the upstream trace. UI Traces offers a contrasting case: modifying stored timestamps preserves action types and order, so the first question is which surviving features predict identity [5]. Refitting that view cannot certify a different downstream one.

When transfer is permitted. The source-local inference result applies when the receiver obtains the tested projection under its evaluated population and condition. A stronger information-theoretic permission has a precise form.

Proposition 1 (protected-only postprocessing). Let S be the hidden target, Z the complete protected representation and Y the downstream observation. If $S \to Z \to Y$ is a Markov chain, including any randomization independent of S given Z, then

$$I(S;Y) \leq I(S;Z).$$

This is the data processing inequality [40, Theorem 2.8.1]: expand $I(S;Z,Y)$ in both orders, use $I(S;Y \mid Z) = 0$ and non-negativity of conditional mutual information. Thus a verified protected-only transformation preserves an applicable mutual-information upper bound. A frontend with additional decoded content or target-dependent state need not satisfy that premise; NetEcho illustrates this second path. The proposition licenses an information bound, not transfer of a particular classifier's accuracy.

5.2. Case 2: choose whether cover may execute

Decision and relevant evidence. WRA provides a privacy-oriented reason to execute cover: create plausible alternative research activity around a user's real request. It issues real and decoy prompts in parallel, returns only the real output, and evaluates interleaved visited-domain sets. With one decoy, its functional-intent reconstruction metric falls from 0.770 to 0.610 [6]. This supports reducing the tested observer's prompt inference. AHB supplies a complementary task-state result: Trip Planning success is 0.75 Raw versus 0.15 for combined Online+Fake humanization, with unintended GUI transitions discussed by the authors [9]. The studies measure different endpoints: WRA tests private-intent inference; AHB tests human–agent distinction and task performance.

What changes after binding. Four execution classes make the choice concrete: (i) transport dummies discarded before dispatch, (ii) requests confined to an isolated shadow environment, (iii) read-only/idempotent operations whose initial effects must be checked, and (iv) state-changing actions. These are analytical classes, not four newly measured defenses. Idempotence constrains repetition; it does not make the first state change safe. WRA establishes a privacy benefit for its domain-view construction, while AHB establishes a state risk for one combined humanization condition. Together they motivate separating *whether cover obscures the target* from *whether its execution preserves the task*. WRA's separate visibility-condition usefulness scores are not paired terminal utility for its decoy experiment.

When transfer is permitted. Retain WRA's inference benefit for its tested domain view and decoy construction. Verified discard or non-effecting isolation excludes direct cover-induced task changes; execution against the real environment instead calls for objective goal and policy checks. Same-condition terminal utility and numerical latency/wire cost remain NR-R for WRA's decoy result. Dangol's live loopback OHTTP/A2A condition separately provides a protocol budget: accuracy 0.30 at about 13× bytes and 11× latency [8]. AHB's non-Fake History+Long point (task success 0.7143; detection 0.9905) further shows that state effects and indistinguishability depend on the configuration. These local findings guide which cover implementation to test, rather than pricing one implementation with another's results.

5.3. Case 3: choose a valid adaptation result and price

Decision and relevant evidence. A service owner choosing cache protection under a workload budget needs both a valid protected-condition attack estimate and a cost for the actual matching mechanism. An adaptation result from another receiving view answers only its own subproblem. UI Traces compares clean- and delay-trained classifiers on the same delayed archive view. That permits the positive source-local conclusion that adaptation recovers substantial prediction under the tested transformation (D2) [5]. SIGLEAK's useful procedural inference under a common SkillGuard5 prompt is a different result, not an installation effect or a paired adaptation contrast [41].

Early Bird's K-specific hit/miss calibration and subsequent KV-oracle simulation are distinct from its GPTCache de-identification prototype's 6ms/4% overhead [3]. KV-Gov measures a 1.3/1.6μs median/p99 hash-based message authentication code (HMAC) increment, while workload-dependent lost cache reuse remains unmeasured; key-equality isolation does not establish isolation of a semantic nearest-neighbor index [23]. SpliceDefense likewise separates its constant-time fusion signal effect from its quantized-padding cost [19]. For full-prompt recovery, Early Bird Table VI also distinguishes queries per recovered token from recovery coverage and all-token cost; the authors already explain this survivor-conditioned denominator.

What changes, and when transfer is permitted. Allow the UI adaptation contrast for its matched target, receiver, transformation and defended test units. Allow each component cost for the measured component and mechanism. A new receiver, workload, combined mechanism or matching primitive needs a corresponding bridge. The original authors supply the measurements; our added output is the explicit choice of which result may inform which decision. Comparing the smallest reported milliseconds or microseconds would answer no common deployment question.

The supplementary archive reanalysis supplies a compact illustration of the same rule: refitting after the specified offline timestamp perturbation improves paired loss by 0.1503 nats on the same defended records (Supplement S4). It illustrates adaptation within a fixed receiving view; the source cases supply the intervention-placement evidence.

5.4. Evidence strength and the resulting decision rule

Table 5 distinguishes direct support, extensions and background. NetEcho anchors the re-emission case. WRA anchors privacy-oriented decoy execution, and AHB anchors the mobile state-effect contrast. UI Traces, Early Bird, KV-Gov and SpliceDefense anchor the adaptation/component case. Each conclusion is tied to the result that measures it; the wider collection supplies context and counterconditions. Supplement S6 reports source-deletion sensitivity.

The practical payoff is choosing the right next measurement: final-hop capture, task-state validation, or mechanism/workload evaluation. The following inventory makes these decisions explicit across result records.

6. Applicable Intervention Evidence and Costs

6.1. Which results answer a deployment question?

All 43 core families have a scoped intervention disposition. Sixteen contain evaluated mitigation components; two use fixed guardrails, four evaluate interventions for other objectives, 13 have no evaluated intervention within the inspected scope, four are diagnostic components, and four have unbound defense assertions. The inventory retains 33 result records across 22 families and 66 utility/cost field positions (Supplement S5).

For deployment comparisons, we select records in which the source deliberately suppresses the studied channel or detectability and measures either that inference outcome or a cost of the mechanism. Fixed common guardrails, ordinary VPN robustness, access blocking, contention noise, signal-generating privacy filters and channel-only calibration remain contextual records. The companion selector reports every inclusion and retained contextual row. The selected records still mix executed inference, offline/simulated evidence and component costs; they have no common denominator for a defense-failure rate.

Table 6 organizes eight defense types by the receiving view and the result available for a deployment decision.

TABLE 5. DEPLOYMENT JUDGMENTS, POSITIVE REUSE CONDITIONS AND APPLICABLE BOUNDARIES. DIRECT EVIDENCE TESTS THE NAMED BRANCH; EXTENSIONS ADDRESS A DIFFERENT MECHANISM, MODE OR COMPONENT.

Decision	Direct support / extension / background	Judgment after binding	Positive license and remaining bridge
Protect the final stream	Direct: NetEcho IV-C–E/Fig.5 (one re-emission source). Extension: UI Traces 6.1/Fig.5, surviving actions. Background: compound-system composition [31].	An upstream-view result does not certify a new content-aware emitter.	Keep the result for the tested view. Verify the same projection, or evaluate the final hop; protected-only postprocessing rules out fresh-content emission, not all leakage.
Implement cover as actions	Direct: WRA 8.1/Table8, private-intent inference under decoys; AHB 5.2.3/Table1/E.3, mobile state effects. Extension: Dangol 9.7 live transport costs.	Decoys have measured privacy value; execution safety is a separate task-state question.	Keep the WRA domain-view inference result. Verify discard/shadow semantics or measure executed goal/policy and resource effects; WRA terminal decoy utility/cost are NR-R.
Choose adaptation and cost	Direct: UI Traces 6.1/Fig.5, matched delayed-view adaptation. Extension: Early Bird V-A/B, KVGov 5.9/7, SpliceDefense VII-C component/mode evidence. Background: protection boundaries [32].	Permit the source-local adaptation effect and component price; withhold a cross-component deployment ranking.	Bind training access and defended test units; match cache-resolution primitive and workload before transferring mechanism or cost.

Combined with the three cases, it identifies what can be used immediately and which additional binding the intended claim requires. Component measurements retain their engineering value even when they do not price an entire service.

6.2. What changes when records, rather than papers, are compared?

We apply the comparison rules to the 33 records, retaining all 528 unordered pairs in the companion audit. The most interpretable subset is the 13 pairs from within the same study family: seven (53.8%) support a source-qualified descriptive comparison, while six (46.2%) concern different primitives, outcomes or cost objects. Being reported in one paper therefore does not make two results one operating point. Of the seven aligned-outcome pairs, four use positively different collection/execution populations and three lack a verified matched-test or fitting binding. Their values cannot be subtracted as paired effects. This leaves intact the valid clean/adapted contrasts *within* the UI Traces and Dangol capture records.

Two examples show the practical change. Dangol's capture and live results describe the same workflow-recovery problem but are different operating points. SpliceDefense's CTBF signal result and QCP cost are useful component measurements, not a measured combined frontier. The audit turns these distinctions into explicit decisions with source locators, rather than a warning based only on terminology. Supplement S8 gives the wider, mechanism-conditioned pair inventory and directional transfer judgments; the counts describe our coded comparisons, not mistakes by the original authors.

Table 7 applies the six reporting checks of Section 8 to the 16 evaluated-mitigation families. Targets, receiving views and placement are usually source-bound; deployment-level joins need more care. AHB supplies detectability and objective task success without a measured execution-cost endpoint. Dangol's live result supplies inference and protocol costs without objective task utility. Speculation couples its inference score to token payload, not a wire-cost or task-success measurement. These complementary results identify the additional measurement needed for the intended deployment, rather than imply that each component study failed its own objective.

6.3. Adaptation, training access and publication runs

Protected-data fitting (D1) and a paired comparison on the same defended test condition (D2) are separate. In the five multifeature-trace subset, UI Traces, NetEcho and Dangol establish D1; UI Traces and Dangol establish D2. Whisper Leak/AHB exact fit bindings remain unresolved, giving source-status bounds of three–five and two–five respectively. These bounds describe this subset, not all learned attacks. Early Bird's K-specific hit/miss calibration is a separate D1 stage before oracle simulation [3].

Labeled protected data must also be obtainable under the observer's permissions. Paired samples, tuning and compute budgets need their own bindings; source-local refitting alone proves neither access nor fairness. For AHB, released code confirms condition-specific refits (D1-code), but the historical Table 1 run (D1-run) and a paired common-test adaptation comparison (D2-paired) remain NR-V [9]. The state-change case uses the separately reported task-success result, not an adaptive-detector effect.

6.4. Costs and excluded quantitative assertions

Interpret cost as the object measured. Dangol's live 0.30 and transformed-capture 0.17 cannot be subtracted as a paired deployment effect. Its 0.167 timing-feature ablation is not an executed timing masker. Early Bird's 6ms/4% semantic prototype does not price KV-granularity protection; KVGov's HMAC microcost omits lost cache reuse. SpliceDefense combines Constant-Time Boundary Fusion

TABLE 6. DEFENSE TYPES AND THE EVIDENCE AVAILABLE FOR CHOOSING THEM. THESE REPRESENTATIVE CONDITIONS SPAN CONFIDENTIALITY, ATTRIBUTION AND HUMAN–AGENT DETECTION; THEIR OUTCOMES ARE NOT A COMMON EFFICACY RANKING.

Type	Evaluated view, target and source	Established result and applicable boundary
Padding / batching	Network packet lengths and groups; topic inference. NetEcho and Whisper Leak [7], [16].	Collected padding traces retain inference signal; packet-group batching is evaluated by replay. A protected upstream length view does not cover downstream re-emission or residual grouping. Live incremental cost remains NR-R.
Constant-rate / transport cover	Network workflow fingerprint; Dangol's OHTTP testbed [8].	The co-located loopback setting yields 0.30 recovery at about $13\times$ bytes and $11\times$ latency. Transformed-capture results are separate; neither result supplies objective agent-task utility.
Concurrent decoy agents	Destination/domain observer; private prompt and trait inference. WRA [6].	Real/decoy requests execute in parallel; interleaved-domain evaluation reduces one-decoy E_{func} from 0.770 to 0.610. Terminal decoy utility/cost are NR-R; report usefulness is a separate visibility condition.
Action humanization	Mobile touch traces; human–agent detection. AHB [9].	Executed combined humanization changes Trip Planning success from 0.75 to 0.15. Fake actions can alter state; offline trajectory edits and other online configurations are separate conditions, with different utility implications.
Cache isolation / keying	Co-tenant prefix timing; SafeKV and KVGov [42], [23].	SafeKV reports post-trigger AUC near 0.5 and separate TTFT/throughput comparisons. KVGov's attack results are simulated; its measured HMAC median $1.3\,\mu$s prices key computation, excluding lost reuse. Each cost retains its mechanism and workload.
Cache timing equalization	Co-tenant cache-fusion timing; SpliceDefense [19].	Constant-Time Boundary Fusion (CTBF) suppresses the tested signal over 15 tasks/75 steps. Quantized Chunk Padding (QCP) adds mean 7.5 tokens and $<$ 5ms/$<$ 1% prefill time; this does not price combined deployment.
Timestamp perturbation	Recorded website action metadata; identity attribution. UI Traces [5].	Fitting on delayed traces recovers substantial predictive performance under offline timestamp edits that retain action types/order. Live execution delay and task outcomes are not measured by the trace transformation.
Oblivious / unshared memory	Host or local resource observations; TDXRay and Gao et al. [43], [44].	TDXRay measures oblivious-map lookup cost; Gao measures loading/memory cost after disabling zero-copy. These are component measurements, not post-mitigation prompt-recovery or complete serving-cost evaluations.

The table summarizes evaluated conditions or measured components, rather than treating proposed mechanisms as deployed defenses. TDXRay's TEE setting specifies the host adversary; it does not establish that TEE residency alone prevents the studied channel. The result-level inventory retains the separate conditions and missing fields.

TABLE 7. SIX CHECKS OVER 16 EVALUATED-MITIGATION FAMILIES. COUNTS APPLY TO THE SOURCE-SCOPED CLAIM; NO COMPOSITE QUALITY SCORE IS COMPUTED.

Check	Bound	Partial	NR-V	NA
1. Target/protection relation	16	0	0	0
2. Receiver fields/permissions	13	3	0	0
3. Placement/action dispatch	13	2	1	0
4. Protected fit/test/access	0	4	5	7
5. Tested workloads/shifts	14	2	0	0
6. Joint inference/utility/cost	0	16	0	0

Bound: the required item is source-bound for the scoped claim. Partial: supported elements coexist with an unbound element. Check4 includes the observer's training-data access and matching, not just existence of a fitted model. Check6 asks for an end-to-end deployment join, not a requirement that every component study perform one. No whole-check cell is solely NR-R; individual unreported subfields remain NR-R in the 96-cell inventory.

(CTBF) and Quantized Chunk Padding (QCP); mean 7.5 padding tokens and $<$ 5ms/$<$ 1% describe QCP alone, not the combined cost [8], [3], [23], [19].

Likewise, Speculation measures $8.7\times$ token payload; Gao measures loading cost and TDXRay map-lookup cost [45], [44], [43]. SafeKV reports both asynchronous detector latency and serving TTFT/throughput measurements; its five-second post-trigger leakage window and serving comparisons retain their own conditions [42]. These are useful measurements of the named objects, rather than interchangeable prices. Supplements S5/S8 preserve the mode-specific bindings and source corrections.

Four assertions remain outside quantitative comparison because their result/run binding is unresolved: AgentPrint's supplementary analysis, Weiss's vendor confirmation, Gu's provider changes and PromptEcho's $3.2\times$/$1.4\times$ padding overhead [4], [1], [46], [47]. The three deployment cases use the other, source-bound records.

7. Match Transfer Claims to the Generator

7.1. Different tests rule out different explanations

A fingerprint is generated by a model interacting with tasks, prompts, tools, browser state and serving infrastructure. A holdout establishes portability along the axis it changes: new tasks, browser configurations, network topologies and hardware are distinct deployment tests. Supplement S2 maps each dependency to its supporting holdout.

UI Traces fixes a browser/action harness while testing tasks and environments. FP-Agent combines behavior with static browser attributes; shared visitor proxies are not authenticated independent people or hosts [5], [25]. Task separation can remove memorization of those task instances while retaining configuration identifiers. This is not proof of an invalid original split; it bounds portability. The traffic benchmark separately tests generating topology, so its shift

is a different test rather than a stronger score on the same estimand [24].

LeakyLMs illustrates the boundary for system-property rather than agent identity. Its deployment probe observes streaming events through an ordinary client API. Its architecture branch additionally assumes known GPU type, same-class calibration and single-GPU execution [22]. Remote architecture inference therefore does not establish hardware-agnostic inference. The joint search in its remote example ranks the true configuration 28/2068, distinct from a conditional parameter search or a near match; those outcomes must not be collapsed into universal exact recovery. Source-reported local architecture generalization and remote deployment signatures remain positive evidence under their own assumptions.

7.2. Representation increment is not portability

Within a fixed observer and test population, coarse counts versus sequence features test whether the latter improve a specified predictor. WRA already uses ordered domains before adding timing; Kang changes class sets across some conditions [6], [48]. These test different contrasts from an order-only effect. The supplementary archive example fixes labels and units and finds a 0.3447-nat sequence increment under archive-receiver ordering. Its small common-task control leaves task-distribution explanations unresolved (Supplement S4).

Adjacent coding-agent PR attribution depends on repositories, submission pathways and artifact generation. Its mixed-feature result does not isolate an auxiliary-only C3 view; inspected code computes weighted F1 and reduces features before cross-validation, but the exact paper-run binding remains unverified [10]. We neither rewrite its headline score nor quantify optimism. WildAuth is adjacent under the same rule; input/partition ambiguities do not supply a model-independent response-only privacy claim [11].

7.3. Resulting claim hierarchy

Task-specific identification, task-disjoint identification in a fixed harness, configuration/topology transfer and hardware-calibrated architecture inference require different evidence. Move between them by changing the relevant generator while preserving target labels and the receiver contract. For a representation increment, compare representations on identical units and disclose predictor/selection restrictions. This claim-to-holdout rule lets a reader reuse each author test for the shift it measures.

8. Implications and Limits

8.1. A deployment evaluation checklist

Before claiming that a defense protects a deployment, report: (1) the hidden target and protection relation; (2) the final receiver's actual fields and linkage permissions; (3) where protection ends and whether cover reaches action dispatch; (4) protected-data access, fitting procedure and a matched defended-test contrast where adaptation is claimed; (5) the workload and the generator shifts actually tested; and (6) inference, objective/policy utility and applicable cost bound to the same condition. Report missing entries as scoped evidence states. The checklist asks for evidence needed by the specific claim, not every measurement in every paper.

The three cases make this actionable. A content-aware downstream renderer calls for paired final-hop captures; surviving actions call for residual-feature prediction; executed dummy actions call for objective state/goal tests. Prefix-cache isolation needs workload reuse costs, while a semantic matcher needs a separate mechanism check. Task, topology, browser configuration and hardware shifts need their respective bridge tests. These follow from the source comparisons rather than from a global defense ranking.

8.2. Research questions implied by the cases

Five concrete questions follow from the decision procedure in Figure 2. (1) Across API, frontend and final receiver, which transformations preserve a protection bound, and which introduce additional target-dependent state? (2) Can a cover policy be certified goal- and policy-neutral before tool dispatch, while accounting for resource and timeout effects? (3) How much protected training data can each real observer obtain, and what is the resulting adaptation frontier under a fixed budget? (4) For isolation or equalization, how do workload reuse and the actual matching primitive change total service cost? (5) Which receiver-visible invariants survive changes in task, topology and backend without relying on private collector fields? These questions prioritize missing bridges over another heterogeneous attack ranking.

8.3. Threats to the synthesis

Selection and source binding. The purposive collection supports source-specific deployment judgments, not literature-wide prevalence or completeness. Missing sources could add counterconditions or closer prior work. Four material/lineage-limited candidates are not admitted, and four unbound defense quantities are withheld from matched comparisons (Supplements S1/S5). Case-level dependence is explicit in Table 5; source deletion withdraws the relevant instance, not the other measured mechanisms.

Interpretive reliability. Eligibility, result splitting and transfer judgments use a single AI-assisted coding workflow. Independent human agreement is unmeasured. The supplied locators, pairwise reasons and alternative dispositions make the interpretations auditable. The retained negative checklist comparison is the reason we position SCOPE as fields, not as a more effective review method.

Empirical reach. The supplementary archive case addresses prediction under archive-receiver ordering; matched-task evidence remains inconclusive and live capture is unverified. Original failed pilots and retail results are retained.

The main contribution instead rests on the cited intervention results and the result-level synthesis.

Disclosure and materials. The companion Supplementary Evidence contains source-linked tables, saved predictions, protocols and a local verifier, excluding private logs, conversations and redistributed third-party datasets. It is prepared for submission as accompanying material; the README provides the entry points. The authors remain responsible for AI-assisted judgments and submission disclosures.

9. Conclusion

The useful unit of defense evidence is an evaluated result tied to a receiving view, an execution boundary and a workload. Organizing the literature at this level yields three actionable decisions: protect the final emitter, validate cover where it executes, and use adaptation effects and costs for the mechanisms actually measured. WRA's privacy benefit, AHB's state effects and the cache components' engineering measurements each retain their value without being forced into one ranking.

The contribution is the bridge from existing measurements to deployment choices. The source-linked cases identify when a result can be reused; the comparability audit records which proposed inferences are supported, mismatched or still need a binding. Together, they turn a diverse side-channel literature into evidence a system designer can apply and a reviewer can reconstruct.

Appendix A.
Reproducibility and Notation

The accompanying *Supplementary Evidence* provides eight self-contained sections: S1 search records and dispositions; S2 corpus, prior-work and receiving-view maps; S3 retained pilots; S4 public-trace reanalysis; S5 intervention inventory; S6 conclusion sensitivity; S7 the original negative workflow comparison; and S8 the pairwise/checklist audit. Its machine-readable tables link family ID → result ID → version and source locator → fields → supported judgment. The README and hash manifest identify the current paper and verifier entry point. Public hosting is separate from this accompanying-file access route.

Executable acquisition record. Table 8 records the actual native queries rather than retroactively improved expressions. IEEE used Command Search/default metadata on October 8, 2026, without lower-year or document-type restrictions, exporting RIS with Citation and Abstract. ACM used the All field in the Full-Text Collection on October 8–9, then E-Publication Date partitions and All Results/BibTeX export. S1 supplies counts, exact partition dates/URLs and retrieval discrepancies. Targeted primary-source discovery and fixed-seed tracing complement these exports; the final analysis is the purposively assembled, version-bound collection.

TABLE 8. EXECUTED NATIVE QUERIES. S1 RETAINS THE FULL EXPORT ACCOUNTING; Q4 WAS ISSUED ONLY ON IEEE AND NOT EXPORTED IN THE ORIGINAL ACQUISITION.

ID	Exact query
Q1	`("large language model" OR LLM) AND ("side channel" OR "side-channel" OR "timing attack")`
Q2	`(agent OR agentic) AND ("traffic analysis" OR fingerprinting OR "metadata leakage" OR "side channel" OR "side-channel")`
Q3	`("KV cache" OR "prompt caching" OR "speculative decoding") AND (privacy OR leakage OR attack)`
Q4	`("browser agent" OR "research agent" OR "tool calling" OR "model context protocol" OR "multi-agent") AND (timing OR fingerprinting OR leakage)`
Q4b	`(("browser agent" OR "research agent" OR "tool calling" OR "model context protocol" OR "multi-agent") AND (timing OR fingerprinting OR leakage) AND ("large language model" OR LLM))`
Q5	`("language model" OR LLM) AND ("traffic shaping" OR "cache isolation" OR padding OR "side-channel defense")`
Q6	`("language model" OR chatbot OR "AI assistant" OR "generative AI") AND ("side channel" OR "side-channel" OR "traffic analysis" OR fingerprinting OR "timing leakage")`

TABLE 9. READING KEY FOR THE SUPPLIED RECORDS. PROSE USES FULL MEANINGS; CODES SUPPORT JOINS.

Notation	Meaning
SCOPE	Secret/target, Channel, Observer, Pipeline, Exploitation goal.
Legacy A/B	Agent-native / runtime/application-inherited signal origins.
C1–C5	Five eligibility conditions in Section 2.
P1–P8	Hardware, serving, transport, application, orchestration, tools, environment, multi-agent communication.
D1 / D2	Protected-condition fitting / matched defended-test comparison of fitting procedures.
NR-R / NR-A	Unreported in inspected scope / unavailable material.
NR-V / NA	Unverified binding / inapplicable.
Q1–Q6, Q4b	Historical search IDs, not quality grades.

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